

PROBLEM STATEMENT

Digital case clues are hard to see together

Notes, network records, and file samples often arrive in different formats.

01**Many input types**

Notes, network traffic, and file fragments are separate kinds of information.

02**Connections get missed**

People, devices, places, and evidence can be difficult to connect by eye.

03**Need one clear view**

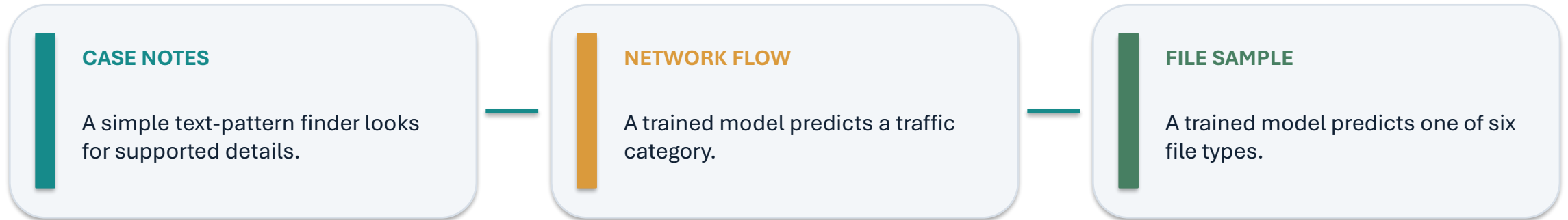
A reviewer needs an organized case view to explore possible links.

Goal: organize sample information and show possible links — not decide guilt.

SOLUTION

Three ways to add sample information

Each workflow adds a result to one case workspace and its relationship graph.



ONE CASE GRAPH

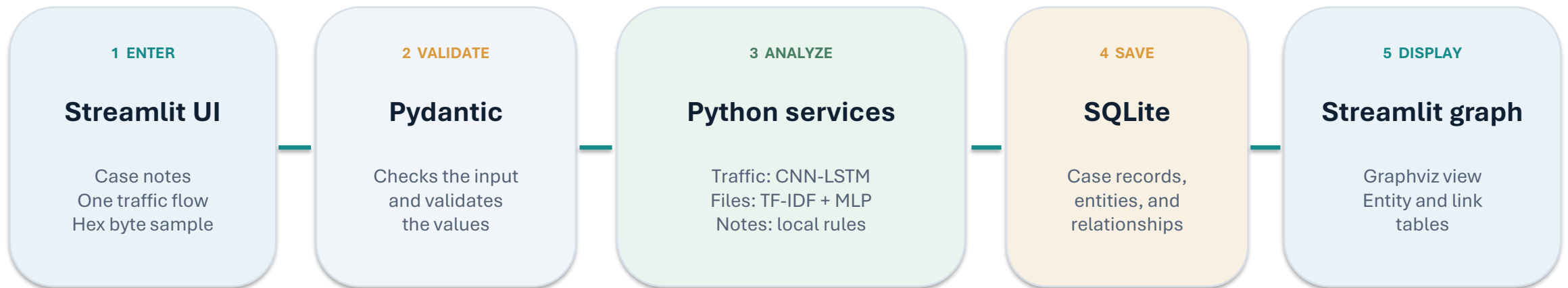
Select a case • view graph • inspect entity and link tables

Two trained classifiers + one separate rule-based notes feature

ARCHITECTURE

Streamlit runs the analysis locally

The interface calls Python services directly; this demo does not send HTTP API requests.



Local Python workflow • no HTTP API calls • results need human review

MODEL COMPARISON

The two classifiers used in the app

A quick comparison of what goes in, what each model does, and what comes out.

MODEL	SAMPLE INPUT	WHAT IT DOES	TRAINING SETUP
Traffic	One flow → 8 × 15 model window	CNN-LSTM predicts benign or malicious traffic	10,000 generated flows 50 epochs
File type	Hexadecimal byte sample	TF-IDF + MLP predicts one of six file categories	6,000 generated samples 50 epochs

Important: the examples are generated by the training code — the original named datasets were not used in the default runs.

RESULTS

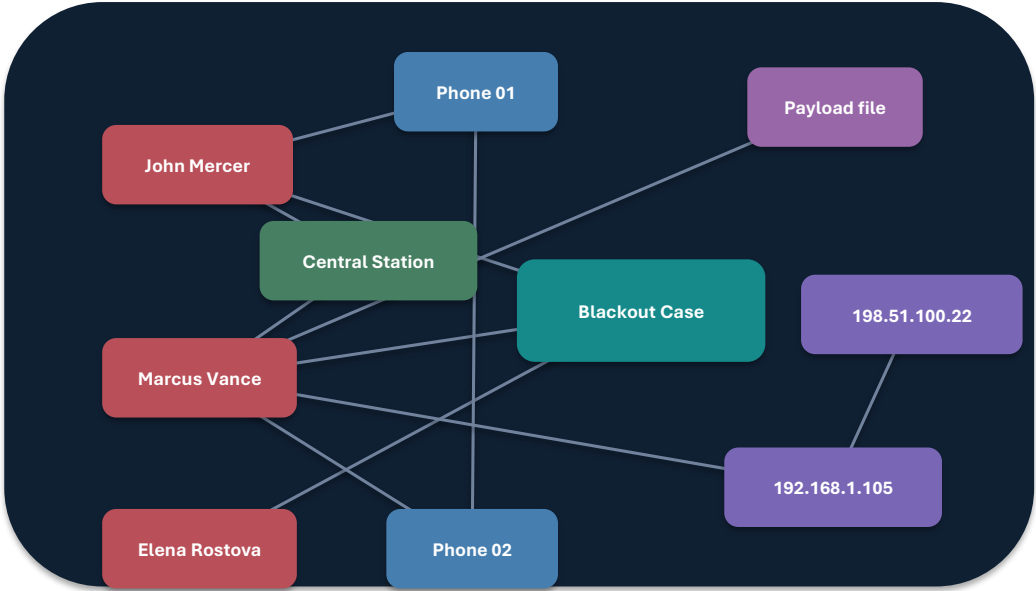
What the prototype produced

Held-out test scores on generated examples — not evidence of real-world accuracy.

MODEL	ACCURACY	F1 SCORE
Traffic CNN-LSTM	98.40%	99.18%
File TF-IDF + MLP	99.92%	99.92%

These scores describe generated test examples only. Real-world performance is unknown.

SAMPLE CASE GRAPH



Fictional seeded case • simplified Streamlit / Graphviz illustration

Next: test on suitable real datasets • independently validate • improve security and storage